

2. Test Objective

2.1 Purpose of the Test

The primary objective of this project is to perform load testing on the BlazeDemo Flight Booking Application using Gatling and evaluate the application's performance, stability, and reliability under simulated user load conditions.

Load testing helps determine how efficiently an application responds when multiple users access the system simultaneously. It enables testers to identify performance bottlenecks, measure response times, analyze throughput, and ensure that the application can handle expected user traffic without failures.

The BlazeDemo application was selected as the Application Under Test (AUT) because it provides a complete flight booking workflow consisting of flight search, flight selection, and booking confirmation processes. These business transactions were used to simulate realistic user activities during test execution.

2.2 Main Objectives

The key objectives of this load testing project are:

1. Evaluate Application Performance

To measure how quickly the BlazeDemo application responds when multiple virtual users perform various flight booking operations simultaneously.

2. Verify Application Stability

To ensure that the application remains stable and responsive during the execution of load tests without producing errors, failures, or unexpected behavior.

3. Measure Response Time

To analyze the time required by the application to process user requests and return responses during different business transactions.

4. Analyze Throughput

To determine the number of requests processed by the application within a specific time period and evaluate the system's ability to handle user traffic efficiently.

5. Monitor Request Success Rate

To verify that all user requests are completed successfully and identify any failed transactions during test execution.

6. Identify Performance Bottlenecks

To detect slow-performing transactions, delays, or resource limitations that may affect the application's performance under load conditions.

7. Validate Business Workflows

To ensure that critical business processes such as flight search, flight selection, and flight booking function correctly under simulated load.

2.3 Load Testing Goals

The following goals were defined for the load testing activity:

- Simulate multiple concurrent users accessing the application.
 - Execute realistic user journeys through the application.
 - Measure application response times for different transactions.
 - Verify that the application handles user requests successfully.
 - Observe application behavior during increasing user load.
 - Generate detailed performance reports using Gatling.
 - Collect performance metrics for analysis and reporting.
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2.4 Scenarios Covered

The load testing process was performed using the following business scenarios:

Scenario 1 – Search Flights

Users access the home page, select departure and destination cities, and search for available flights.

Scenario 2 – Choose Flight

Users search for flights and select a flight from the available options.

Scenario 3 – Complete Booking

Users complete the entire booking process by searching flights, selecting a flight, entering passenger details, and confirming the booking.

These scenarios represent the most important functionalities of the BlazeDemo application and provide a realistic representation of user activity.

2.5 Expected Outcomes

The expected outcomes of the load testing activity are:

- Successful execution of all test scenarios.
- Accurate measurement of response times and throughput.
- Identification of any performance issues or bottlenecks.
- Verification of application stability under load.
- Generation of detailed Gatling performance reports.
- Collection of data required for performance analysis and project reporting.

The results obtained from these tests help assess the overall performance and reliability of the BlazeDemo Flight Booking Application and provide valuable insights for future optimization and scalability improvements.

This section will typically take **1.5–2 pages** in your PDF and is detailed enough for an academic project.